

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **10**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 10**

Find the first four terms of the sequence given by the following.

$$a_n = 61 - 3(n - 1), \quad n = 1, 2, 3 \dots$$

_____, _____, _____, _____

Question 2 of 10

Find the first four terms of the sequence given by the following.

$$a_n = 6 \cdot (-5)^{n-1}, \quad n = 1, 2, 3 \dots$$

_____, _____, _____, _____

Question 3 of 10

Evaluate.

$$\frac{5!7!}{3!9!}$$

Simplify your answer as much as possible.

Question 4 of 10

For each sequence, determine whether it appears to be arithmetic.
If it does, find the common difference.

4, 7, 11, 14, ...	☐ Arithmetic Common difference: $d =$ ☐ Not arithmetic
2, - 8, 32, - 128, ...	☐ Arithmetic Common difference: $d =$ ☐ Not arithmetic
-8, - 11, - 14, - 17, ...	☐ Arithmetic Common difference: $d =$ ☐ Not arithmetic

Question 5 of 10

For each sequence, determine whether it appears to be arithmetic, geometric, or neither.

2, 1, 1, 2, ...	☐ Arithmetic ☐ Geometric ☐ Neither
2000, 200, 20, 2, ...	☐ Arithmetic ☐ Geometric ☐ Neither
13, 15, 17, 19, ...	☐ Arithmetic ☐ Geometric ☐ Neither

Question 6 of 10

For each sequence, determine whether it appears to be geometric.
If it does, find the common ratio.

(a) 4, 20, 100, 500, ...

Classify the sequence and find the common ratio (if any).

☐ Geometric

Common ratio: $r =$ _____

☐ Not geometric

(b) 3, 6, 9, 12, ...

Classify the sequence and find the common ratio (if any).

☐ Geometric

Common ratio: $r =$ _____

☐ Not geometric

(c) 256, -64, 16, -4, ...

Classify the sequence and find the common ratio (if any).

☐ Geometric

Common ratio: $r =$ _____

☐ Not geometric

Question 7 of 10

Find the sum.

$$\sum_{i=1}^4 (2i + 1)$$

$$\sum_{i=1}^4 (2i + 1) = \underline{\hspace{2cm}}$$

Question 8 of 10

Find the sum.

$$\sum_{n=1}^4 (-2n^3)$$

$$\sum_{n=1}^4 (-2n^3) = \underline{\hspace{2cm}}$$

Question 9 of 10

Find the sum.

$$\sum_{j=2}^4 \left(\frac{1}{3}\right)^j$$

$$\sum_{j=2}^4 \left(\frac{1}{3}\right)^j = \underline{\hspace{2cm}}$$

Question 10 of 10

Compute each sum below.

Give exact values, not decimal approximations.
If the sum does not exist, write "Does not exist".

$$\left(-\frac{4}{9}\right) + \left(-\frac{4}{9}\right)^2 + \left(-\frac{4}{9}\right)^3 + \dots = \underline{\hspace{2cm}}$$

$$\sum_{j=1}^{\infty} (-2)^j = \underline{\hspace{2cm}}$$

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Question 1 of 10

61, 58, 55, 52

Note that $a_n = 61 - 3(n - 1)$ is an arithmetic sequence.

The first term of the sequence is 61.

Then, each term is 3 less than the term before it.

Question 2 of 10

6, -30, 150, -750

Note that $a_n = 6 \cdot (-5)^{n-1}$ is a geometric sequence.

The first term of the sequence is 6.

Then, each term is -5 times the term before it.

Question 3 of 10

$\frac{5}{18}$

Question 4 of 10

4, 7, 11, 14, ...	☐ Arithmetic Common difference: $d =$ ☒ Not arithmetic
2, -8, 32, -128, ...	☐ Arithmetic Common difference: $d =$ ☒ Not arithmetic
-8, -11, -14, -17, ...	☒ Arithmetic Common difference: $d = -3$ ☐ Not arithmetic

Question 5 of 10

2, 1, 1, 2, ...	☐ Arithmetic ☐ Geometric ☒ Neither
2000, 200, 20, 2, ...	☐ Arithmetic ☒ Geometric ☐ Neither
13, 15, 17, 19, ...	☒ Arithmetic ☐ Geometric ☐ Neither

Question 6 of 10

(a) ☒ Geometric

Common ratio: $r = 5$

☐ Not geometric

(b) ☐ Geometric

Common ratio: $r = \square$

☒ Not geometric

(c) ☒ Geometric

Common ratio: $r = -\frac{1}{4}$

☐ Not geometric

Question 7 of 10

$$\sum_{i=1}^4 (2i + 1) = 24$$

Question 8 of 10

$$\sum_{n=1}^4 (-2n^3) = -200$$

Question 9 of 10

$$\sum_{j=2}^4 \left(\frac{1}{3}\right)^j = \frac{13}{81}$$

Question 10 of 10

$$\left(-\frac{4}{9}\right) + \left(-\frac{4}{9}\right)^2 + \left(-\frac{4}{9}\right)^3 + \dots = -\frac{4}{13}$$

$$\sum_{j=1}^{\infty} (-2)^j \text{ Undefined}$$